



DHIRUBHAI AMBANI GREEN ENERGY
GIGA COMPLEX, JAMNAGAR,
GUJARAT, INDIA



Project Title

INTEGRATED PV MANUFACTURING FACILITY- INGOT WAFER PILOT PLANT

DOCUMENT TITLE:

TECHNICAL SPECIFICATIONS FOR HVAC LOW SIDE SYSTEM

DOCUMENT NUMBER:

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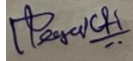
					
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Table of Contents

Table of Contents	3
1. Introduction	5
2. General Requirements	6
3. Scope of work	7
4. Applicable Code and Standards	8
5. Conflicts & Deviations	9
6. Basis of Design	10
6.1 Site Information	10
6.2 Utilities	10
6.3 Ambient Condition	10
6.4 Inside Conditions	11
6.5 Internal Equipment Loads	11
6.6 Lighting Load	11
6.7 Human Occupancy	11
6.8 Fresh Air	11
6.9 Overall Heat Transfer Coefficient	12
6.10 Room Differential Pressure	12
6.11 Noise Criteria	12
6.12 Battery Limits	13
6.13 HVAC Zoning	13
7. Brief HVAC System Description	14
7.1 Puller Side	14
7.2 Wafer Side	14
7.3 Office Area	15
7.4 Electrical Room	15
7.5 SCADA/IT/SERVER Room	16
7.6 Battery Room	16
7.7 Low Side System	16
8. Technical Specifications:	17
8.1 Standard RIL Specifications for Various Equipment/Components.	17
8.2 Vibration Isolation	17
8.3 Motorized Fire and Smoke Damper	18
8.4 Chilled water Manifold Piping	19
8.5 Valves	21
8.6 Insulation For Chilled Water Pipe and Ducting	21
8.7 Electrical Cables	21
8.8 Electrical Panel	21
8.9 VFD	21

	DHIRUBHAI AMBANI GREEN ENERGY GIGA COMPLEX, JAMNAGAR, GUJARAT, INDIA	
---	---	---

8.10	Filters	22
8.11	Interlocking.....	22
8.12	Axial Fans	22
8.13	Motor Name Plate Requirement	23
9.	System Design Parameters	25
10.	Smoke Extraction System	26
11.	Pressurization System for Fire Escape Corridor	27
12.	Integration Between HVAC and Building Automation System (BAS)	28
12.1	AHU	28
12.2	DEC	28
12.3	VRF	29
12.4	General Points.....	29
13.	Vendor Document Requirement	30
14.	Acceptance Test	31
15.	Assumptions/ other Considerations/ Assumptions/ Calibration Required	32
16.	Energy.....	33
17.	Safety	34
18.	Abbreviations:.....	35

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1. Introduction

As a part of the “Carbon Neutral Economy” initiatives, RIL is planning to set-up an integrated “Solar PV Manufacturing” unit in Dhirubhai Ambani Green Energy Giga Complex (DAGEGC), Jamnagar. PV manufacturing facilities shall have end to end integrated facilities for the entire solar value chain (Mg-Si, Poly-Si, Ingot Wafer, Cell, Module and Glass) along with all ancillary facilities. The production of Ingot and Wafer from Polysilicon is one among the entire PV manufacturing value chain. The Pilot Plant is being set-up for a capacity of 130MW and will be used for training RIL’s operating personnel. RIL intends to dismantle this set-up eventually and get the equipment relocated to the main plant once it becomes operational.

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2. General Requirements

- A. The vendor shall make all possible efforts to comply strictly with the requirements of this specifications and other specifications / data sheets as mentioned in the inquiry/ order. In case, any deviations are considered essential by the vendor after making all possible efforts, these shall be separately listed in the vendor's offer under "List of deviations / exceptions to the inquiry" for the purchaser's considerations. The deviations / exceptions shall be listed separately for each specification / document with cross-references and proper reasons for the deviations / exceptions. In the event of any deviation not listed under the 'List of deviations / exceptions to the inquiry', even though it is appearing in other part of the offer, shall not be considered applicable. Hence all the requirements shall be binding on the vendor without any cost and schedule implications to the purchaser.
- B. Compliance to this specification shall not relieve the vendor of the responsibility of supplying the equipment and auxiliaries of proper design, material, and workmanship to meet the operating requirements specified in the Datasheet. In case of requirements of additional instruments, controls, safety devices and any other auxiliaries or accessories for safe, smooth, and trouble-free operation of the equipment; vendor shall include the same in his scope of supply and give reasons with the offer.
- C. The equipment supplier shall be responsible for the co-ordination of all Sub-Suppliers and for the overall guarantee of furnished equipment. It is the specific responsibility of the Supplier to invoke all applicable referenced specifications to each Sub-Supplier Purchase Order.
- D. The offered equipment shall be sourced from a regular and established manufacturer and having requisite design, engineering, manufacturing, and testing facility. The offered model shall be from the regular manufacturing range of the supplier and the design shall have had a minimum of two years of successful field operation with the same model having the same design and materials of constructions. Relevant information of the offered equipment like catalogues, references, user's certificates etc. shall be furnished along with the offer.

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3. Scope of work

The scope covers design, engineering, detailed shop drawing, manufacturing, inspection, transportation, unloading / handling, erection, testing and commissioning of HVAC Low Side system complete with all equipment, accessories.

It is not the intent to completely specify all details of design and construction for the HVAC systems under consideration. The equipment and workmanship at site must conform to high standards of design, engineering, and skill. All items of work, which are required to make the system complete, shall be deemed to have been included in their offer. Offered Equipment Shall be as per Specified in MR and after award of Contract Contractor shall do their own Heat Load/Ventilation calculations and take Approval form DEC/RIL and offer the equipment accordingly to meet the specified inside conditions in all seasons. The complete facility shall be designed as per standard engineering practice and ISHRAE standards.

The Contractor must submit the technical data sheets and provide documentation requirements as given in 'Vendor Document Requirements'.

Detailed Design will be provided by DEC/Jacobs and based on the input provided, vendor to validate the design, it will be the responsibility of the Contractor to coordinate with the Client/Consultant and demonstrate the system performance, as per the intent of this contract.

The work under this contract shall consist of the supply and execution of a complete HVAC Low Side System as per drawings and specifications including but not limited to the following:

- Supply & Installation of HVAC Low Side System
- Transportation to site
- Unloading at the site
- All electrical/on-site coordination with services consultants and site contractors.
- Testing, commissioning at the site, and handing over.
- Maintenance during the warranty period shall be as specified in the commercial terms and Conditions.
- All submittals as per specifications

For details of the scope matrix, refer to Annexure-IV enclosed herewith.

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4. Applicable Code and Standards

The following Standards / Guidelines shall be followed unless otherwise specified, for other Codes which not mentioned here vendor to follow respective Package Specification/RPMS Specification:

Codes	Description
IS:4894-1987 (Reaffirmed 2014)	Centrifugal fans
NFPA 90A & 90B-2018	Standard for the Installation of Air Conditioning and Ventilating Systems
SMACNA	Metal air ducts fabrication/testing
IS:325-1996(Reaffirmed 2012)	Three phase induction motors
IS:1231-1974 (Reaffirmed 2013)	Dimension of 3 phase foot mounted induction motors
IS 4691-1985 (Reaffirmed 2009)	Degrees of protection provided by enclosures rotating electrical machinery
IS: 2062-2011	Steel for general structural purposes
ISO 16890:2016	Air filters for general ventilation
IS 3315	Evaporative Air Coolers
AMCA Standard 210	Air Movement & Control Association Standard 210 for laboratory method of testing fans for aerodynamic performance rating
Safety standard:	
IS:659 1964(R2017)	Safety code for Air conditioning
Central Electricity Authority	Measures relating to safety & electric supply Regulations 2019
OSHA	Occupational Safety and Health Administration
Material standard:	
IS:1239-1990 Part I & II/IS 3589	Mild steel tubes, tubulars & other wrought steel fittings
IS 277-2003	Galvanized sheet steel

Any code that is not specified above but is mandatory for HVAC system design or is a local regulation shall also be applicable.

Note:

1. The latest version of all the standards should be referred

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5. Conflicts & Deviations

In the event of conflict or deviation from requirements stipulated in this specification and other applicable documents, the following order of precedence shall govern:

1. Data Sheets and Drawing
2. The Material Requisition
3. Standard Specification
4. Codes and Standards Specifically Referred
5. Other National and International Standards as Referenced

Latest revisions of codes and other statutory requirements as applicable at the time of enquiry, shall be applicable. In the event of a conflict between the requirements of the referenced codes, standards, specifications, regulations, and procedures, the Supplier shall submit the recommended solution to the Buyer for review and permission to proceed before implementation.

Notes:

- “Data Sheets and Drawings” represent those attached to the Material Requisition.
- “Material Requisition” represents Material Requisition for Quotation, superseded by Material Requisition for Purchase on execution of a Purchase Order.

Any local statutory and regulatory requirement shall mandatorily be applicable and addressed during the engineering stage.

6. Basis of Design

This specification is to define minimum technical requirements for design, materials, manufacture, inspection, testing, packing, shipment, and documentation HVAC Low Side System, for the supplier and/or Contractor to comply with, to achieve the Owner's objective of standardization.

6.1 Site Information

- a) Location of Site: Jamnagar, Gujarat.
- b) Latitude & Longitude: 22° 39' N & 69° 50' E.
- c) Altitude: 20 meters above sea level
- d) Hazardous Classification : No
- e) Seismic Zone – Zone IV as per IS.1893 -1894.

6.2 Utilities

Electricity:

Power : 415V $\pm$ 10%, 50 Hz $\pm$ 5%, 3 Phase, 5 wire.

Control voltage : 230 V AC (or 220V AC), 50Hz, 1Ph. to be derived by Contractor from

415V, either by providing suitable control transformer or direct tapping from phase & neutral.

6.3 Ambient Condition

- a) JAMNAGAR SITE INFORMATION

	Summer	Monsoon	Winter
DBT (°F) / (°C)	100.4 / 38	84.9 / 29.4	55.0 / 12.8
WBT (°F) / (°C)	80.06 / 26.7	80.96 / 27.2	48.92 / 9.4
(RH)	42%	85%	65%
Humidity Ratio (Gr/lb.)	123.32	154.6	41.65

- b) Summer dry bulb temperature daily range: 11.1 °C
- c) Buildings exposure: Solar PV Factory
- d) Region: Coastal

However, the equipment shall be selected to operate at 43°C without tripping and without Deration. Only Critical Equipment shall be capable of running at Jamnagar at 48 + 2 Deg C without tripping.

6.4 Inside Conditions

Kindly Consider Room wise Parameter list Attached as Annexure.

6.5 Internal Equipment Loads

Heat dissipation data from the equipment to room is provided as per FUM data and same will be considered for the heat load estimations.

6.6 Lighting Load

Lighting Load shall be considered based on Electrical team inputs and as per ASHRAE 90.1.

6.7 Human Occupancy

Light bench work activity level needs to consider for estimating heat load generated by occupant for Process Area and for Comfort Area follow below table.

	Heat Gain from occupancy (Office work)
Heat Gain	Metabolic Rate (Btu/hr.)
Sensible Heat gain	245
Latent Heat gain	205

Maximum no. of people working at any time will be taken while designing the HVAC system.

For initial calculation, occupancy density will be as per ASHRAE 62.1 guideline. The occupancy of individual room shall be taken from individual discipline owner and the same will be considered for calculation during detailed engineering.

For Air Conditioning Area need to consider occupancy as per seating arrangement shown in layout.

Electrical Rooms and Storage rooms Assumed minimum occupancy and corridors with No occupancy.

6.8 Fresh Air

Fresh air for Air-conditioned areas

In production /Manufacturing areas / Office Areas:

- As per ASHRAE 62.1-2016
- Exhaust air from fume hoods / dust collector / equipment shall considered as per the FUM data.
- For calculations of leakages due to pressure difference between two areas, the following gaps shall be assumed for the doors,

- Leaf type door (Single Leaf/Double leaf)

Bottom - 3mm

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Sides & Top - 1mm / 2mm as applicable

Centre (Double leaf doors) - 1mm

- Sliding type door (If applicable)
- Periphery - 5mm.

Fresh air for Non-AC (Ventilation areas)

- Air changes for ventilated (non- air conditioned) areas will be considered as mentioned below as per NBC/ local Authority.

Toilets/Washrooms

minimum 06 to 10 ACPH or 75 CFM per unit whichever is higher.

Utility areas

minimum 15 to 30 ACPH

Switchgear Room/MCC Room

Airflow Based Evaporative Cooling Calculations as per Heat Dissipation in Electrical Rooms for temp MNT 40 Deg C.

Ingot & Wafer Area

Note 1-Based on heat dissipation and temp raise and other factor the final ACPH will be arrived for Ventilation and Evaporative Cooling.

Note 2-Minimum ventilation required in factory building shall be provided to meet local state factory act / NBC-2016 requirement.

6.9 Overall Heat Transfer Coefficient

Factor for Heat transfer Coefficient (Uvalue) shall depend upon material of construction of walls and roofs. Same shall be finalize and calculated during detail engineering and will be part of Engineering Calculations.

6.10 Room Differential Pressure

Differential pressure in rooms shall be positive with respect to ambient. Room with hazardous area classification shall be negative with respect to non-hazardous room but positive with respect to ambient. Refer attached Annexure A for detailed information.

(Note: - Ingot Area Shall be Considered Positive with reference to adjacent Area)

6.11 Noise Criteria

Fans shall be mounted on vibration isolators. Air handlers shall be mounted on rubber pads or mounts according to load capacities and expected deflections. Kindly refer RPMS Specification of respective Equipment for recommended Noise criteria.

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6.12 Battery Limits

All utility required for HVAC system shall be provided as below.

- **Electrical Power Supply**

Single Point Electrical power supply shall be provided up to HVAC Starter / VFD panel. Further distribution of Power, Cable, earthing, cable trays etc shall be done by HVAC contractor / Client.

- **Chilled Water Supply**

Chilled water supply shall be provided up to HVAC equipment manifold based on HVAC Equipment Layout. Chilled water inlet temperature shall be 7°C & outlet shall be 12°C. Chiller, pump & all associated piping up to HVAC equipment manifold shall done by others.

VRF system Shall be considered as option.

- **RO Water**

RO supply shall be provided up to HVAC unit further connection shall be done HVAC contractor.

- **Condensate / Drain**

All HVAC/AHU room shall have HVAC drain trap and HVAC condensate drainpipe will be connected up-to HVAC room drain trap (Further distribution by other) and drainpipe should not be connected to any drain having foul smell building drains.

- **BMS of HVAC**

Field Devices, DDC controller, required software with system for HVAC Control shall be provided by HVAC Contractor.

NOTE

For detailed scope understanding kindly refer **Annexure I – Scope Matrix**

6.13 HVAC Zoning

Various areas would be segregated into Zones to determine the number of HVAC systems (AHUs/DEC/VUs/EUs) requirement based on following main parameters:

- Activity
- Process parameters
- Heat gain / Load distribution
- Operational procedures
- Building layout

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7. Brief HVAC System Description

7.1 Puller Side

Process area like Puller area First Floor & Hot Zone, Hot Cleaning & Poly charging Room, Polysilicon Storage/Quartz Crucible Unpacking at ground floor will be Cater by Direct Evaporating Cooling which shall maintain Temperature for Human Comfort around 30°C, at spot location for occupied/operational zone and inside area shall be complied as per Gujrat Factory act requirement. (Please refer Annex. A for Area wise Details)

Direct Evaporative Cooling unit shall be installed in HVAC Technical Area at First Floor, DEC unit shall comprise of Fresh air section, Pre Filter (G4) and Provision for Fine Filter (F7), Supply air DIDW Belt driven Fan with TEFC Induction motor, Mist Eliminator, Supply air damper ,overflow, drain etc as per Approved datasheet Requirement.

Supply Air from DEC for First floor Puller area shall be supplied through Ducting which run above EOT Crain and dropped down by means of Column support up to 1 meter above Floor and Air shall be supplied by means of Grille/Diffuser to maintain temp for Human Comfort at spot location for occupied/operational zone.

Supply Air from DEC for Ground floor Puller area shall be supplied through Ducting which run inside the room and dropped down by means of Column support up to 1 meter below Floor and Air shall be supplied by means of Grille/Diffuser to maintain temp for Human Comfort.

Exhaust Air shall be exhausted by means of louvers placed on Periphery of building and at ceiling level for transfer air to adjacent Area as per ventilation requirement.

Refer HVAC layouts and Airflow Diagrams.

7.2 Wafer Side

Process area like wire sawing, Demount Room, Wire sawing storage, Shaping Room, Recycle & Fresh Poly Charger Room, Final Cleaning/Wafer inspection, will be Cater by Direct Evaporating Cooling which shall maintain Temperature for Human Comfort around 30°C, at spot location for occupied/operational zone and inside area shall be complied as per Gujrat Factory act requirement. (Please refer Annex. A for Area wise Details)

Direct Evaporative Cooling unit shall be installed in HVAC Technical Area at First Floor, DEC unit shall comprise of Fresh air section, Pre Filter (G4) and Provision for Fine Filter (F7), Supply air DIDW Belt driven Fan with TEFC Induction motor, Mist Eliminator, Supply air damper ,overflow, drain etc as per Approved datasheet Requirement.

Supply Air from DEC for Ground floor wafer area shall be supplied through Ducting which run above EOT Crain (where Applicable) and dropped down by means of Column support up to 1 meter Above Floor and Air shall be supplied by means of Grille/Diffuser to maintain temp for Human Comfort.

Exhaust Air shall be exhausted by means of louvers placed on Periphery of building and at ceiling level for transfer air to adjacent Area as per ventilation requirement.

Rest area of wafer side like Ingot Cooling Area, PCW Package, DI package, PCH Package, Transfer station, Central shall be cater by means of Transfer Air / Ventilation system.

Refer HVAC layouts and Airflow Diagrams.

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7.3 Office Area

Offices, meeting rooms, VIP lounge, Utility monitor, canteen, Supervisor Room, WGR, Slotting, Technology Lab. Instrument Operator room & Spare part office shall be catered by Air Handling Unit.

VRF system Shall be considered as option for Admin Area.

Refer RPMS Design Basis for HVAC systems RPMS-ENGG-DBD-HV-001

Building / Room	Internal Design Temp	Relative Humidity, %	Room Pressure (with respect to adjoining area)
Offices, Admin, Guest house	24 ± 2	<65	Neutral
Toilets, Washrooms, Locker Rooms	Ambient	No Control	- VE
Dining Area,	24 ± 2	<65	+VE

7.4 Electrical Room

Electrical Room will be catered by Direct Evaporating Cooling Unit.

Refer RPMS Design Basis for HVAC systems RPMS-ENGG-DBD-HV-001

Building / Room	Internal Design Temp	Relative Humidity, %	Room Pressure (with respect to adjoining area)
Electrical Switch Gear / Panel Room	Not to exceed 40°C	Non-Condensing	+VE
IT / Electronic	22 ± 1	50% ± 5	+VE
Battery Room	27 ± 2	No Control	-VE (Note1)

Note: Subject to entry door of Battery room is not directly exposed to ambient

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7.5 SCADA/IT/SERVER Room

SCADA/IT Cabinet/Server/UPS Room shall be provided with CHW based Fan Coil unit (FCU). HVAC Unit shall be selected for 24 x 7 Application.

7.6 Battery Room

Battery Room shall be provided with CHW based Fan Coil unit (FCU) which shall be kept in Room to maintain temp requirement 27 ± 2 C. HVAC Unit shall be selected for 24 x 7 Application.

Exhaust Fan shall be provided to exhaust the air from the room to maintain hydrogen level.

7.7 Low Side System

HVAC design is being carried out based on Equipment load / Heat dissipations, Occupancy, Light Load, Exhaust Air, Fresh Air etc. Filtered, cooled, and dehumidified air from the respective AHUs/DEC shall be fed to the air conditioning space of that zone by means of insulated / Non insulated GI/FRP ducting for lab Area (if required).

All AHUs / DEC / Fans etc for Pilot block shall be placed on First floor technical Area.

Washrooms, dirty rooms etc shall be ventilated / exhausted using inline exhausts fans.

Each AHU/DEC shall cater to specified area and the same will be explained in the HVAC zoning drawing

8. Technical Specifications:

8.1 Standard RIL Specifications for Various Equipment/Components.

RPMS STANDARD DOCUMENTS	DOCUMENT NOS
Air Handling Units	RPMS-ENGG-SPC-HV-008
Fan Coil units (FCU)	RPMS-ENGG-SPC-HV-015
Ventilation fans	RPMS-ENGG-SPC-HV-013
Variable Refrigerant Flow System	RPMS-ENGG-SPC-HV-009
Air distribution system & Ducting	RPMS-ENGG-SPC-HV-018
Thermal insulation	RPMS-ENGG-SPC-HV-019
Precision Air conditioner Unit	RPMS-ENGG-SPC-HV-010
Packaged Air conditioner Unit	RPMS-ENGG-SPC-HV-011
Design Basis for Piping Supports	RPMS-ENGG-DBD-ST-003
Standard Specifications for Piping	RPMS-ENGG-SPC-PP-038-PART-I
Standard Specification for LV Switchgear	RPMS-ENGG-SPC-EL-002
Standard Specification for LV Motor	RPMS-ENGG-SPC-EL-004
Standard Specification for Low Voltage Variable Speed Drives	RPMS-ENGG-SPC-EL-011
Standard Specification for LV Power and Control Cables	RPMS-ENGG-SPC-EL-022
Specification for Electrical requirements for Packaged Mechanical Equipment	RPMS-ENGG-SPC-EL-035
LV MOTOR	RPMS-PROC-ITP-EL-001

NOTE

For other Specifications kindly refer MR Annexures

8.2 Vibration Isolation

Special attention shall be given to ensure that vibration isolator to be designed for Seismic Zone IV requirement and there is no transmission of objectionable vibration to the building or occupied space. Vibration isolation shall be provided as required for equipment, duct work and piping as required.

Vibration isolators shall be selected in accordance with the weight distribution and the manufacturer's recommendation to produce reasonably uniform deflection.

8.3 Motorized Fire and Smoke Damper

Motorized Fire Damper-

Fire Dampers & Smoke and fire dampers should meet the requirements of NFPA90A, 92A & 92B. Fire resistance rating certification shall be UL555 for Fire Endurance, Operational Reliability, Dynamic Closure Test. Fire resistance rating shall be a 90 minute. Switch to be provided for damper blade position indication. Sleeve required as per UL/NFPA-standard. Actuator shall be mounted externally on outside of the sleeve. Sleeve length be extended enough to install actuator and retaining angle as per UL 555S Requirement. Sleeve gauge shall be 16G up-to 900 mm duct width and over 900 mm width 14 Gauge. Each fire & smoke damper shall be AMCA licensed with proper seal of air performance.

Damper frame shall be roll formed structural hot channel, reinforced at the corner, formed from a single piece 16g GSS. Blades shall be single skin, 16G GSS with three longitudinal 'V' groove for proper reinforcement. Bearings shall be stainless steel turning in an extruded hole in the frame. Blade edge seals shall be silicon rubber and galvanized steel mechanically locked into the blade edge. Each damper shall be supplied with factory sleeve. The manufacturer must furnish complete GA drawing and obtain approval before manufacturing.

Combined Smoke & Fire Damper-

Fire Dampers & Smoke and fire dampers should meet the requirements of NFPA90A, 92A & 92B. Fire resistance rating certification shall be UL555 for Fire Endurance, Operational Reliability, Dynamic Closure Test and & 555S for leakage class. CSFD shall have factory mounted appropriate actuator with UL label. Fire resistance rating shall be a 90 minute. Actuator shall be mounted externally on outside of the sleeve. Sleeve required as per UL/NFPA-standard. Actuator shall be mounted externally on outside of the sleeve. Sleeve length be extended enough to install actuator and retaining angle as per UL 555S Requirement. Sleeve gauge shall be 16G up-to 900 mm duct width and over 900 mm width 14 Gauge. Each fire & smoke damper shall be AMCA licensed with proper seal of air performance. In addition, the dampers and its actuators shall have a UL555S temperature rating of 250°F.

Damper frame shall be roll formed structural hot channel, reinforced at the corner, formed from a single piece 16G GSS. Blades shall be single skin, 16G GSS with three longitudinal 'V' groove for proper reinforcement. Bearings shall be stainless steel turning in an extruded hole in the frame. Blade edge seals shall be silicon rubber and galvanized steel mechanically locked into the blade edge. Each damper shall be supplied with factory sleeve. The manufacturer must furnish complete GA drawing and obtain approval before manufacturing.

Fusible linked Fire damper-

Fire Dampers should meet the requirements of NFPA90A, 92A & 92B. Fire resistance rating certification shall be UL555 for Fire Endurance, Operational Reliability, Dynamic Closure Test. Fire resistance rating shall be a 90 minute. Switch to be provided for damper blade position indication. Fusible link rating shall be 165°F. Sleeve required as per UL/NFPA-standard. Minimum sleeve length extend beyond wall shall be 150 mm or Sleeve length extend beyond wall shall be extended enough to install actuator and retaining angle in case of the actuator spring return mechanism as Per UL 555S requirement. Each fire damper shall be AMCA licensed with proper seal of air performance.

Damper frame shall be roll formed structural hot channel, reinforced at the corner, formed from a single piece 16G GSS. Blades shall be single skin, 16G GSS with three longitudinal 'V' groove for proper reinforcement. Bearings shall be stainless steel turning in an extruded hole in the frame. Blade edge seals shall be silicon rubber and galvanized steel mechanically locked into the blade edge. Each damper shall be supplied with factory sleeve. The manufacturer must furnish complete GA drawing and obtain approval before manufacturing.

8.4 Chilled water Manifold Piping

The Chilled water pipe will be in CS A 106 Gr. B/ IS 1239/ IS3589

Piping Support Design Reference Document: - RPMS-ENGG-DBD-ST-003

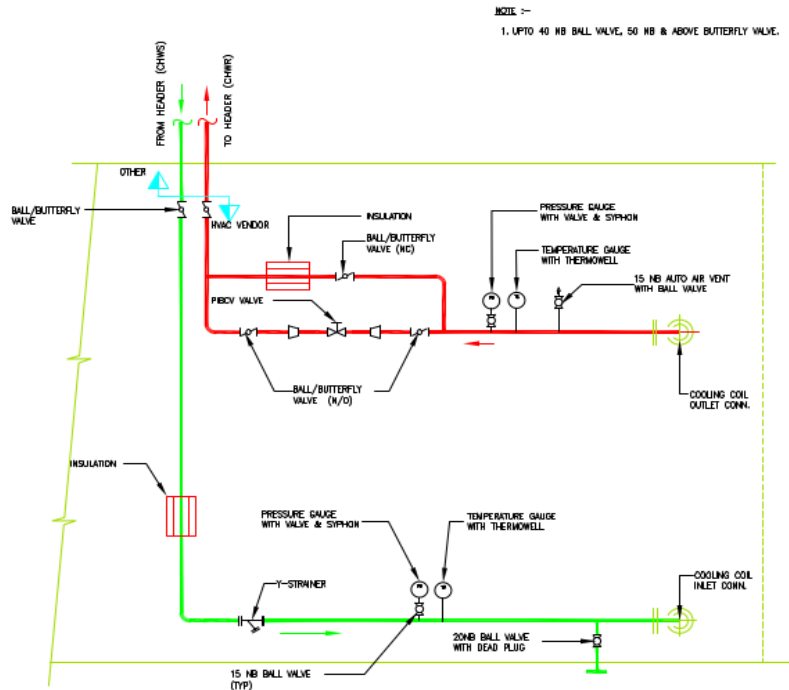
Piping Specification RPMS Document No. J22N03-SPC-PA-820-001 _Rev 0A And RPMS-ENGG-SPC-PP-038-PART-I

Hydro-testing will be done at 1.5-time design pressure.

The main supply and return chilled water header shall be provided by mechanical/piping vendor and further tapping to each AHU including manifold shall be in scope of HVAC vendor Chilled water manifold shall have following valve & accessories for 7-12 Deg.C Chilled water Coil: -

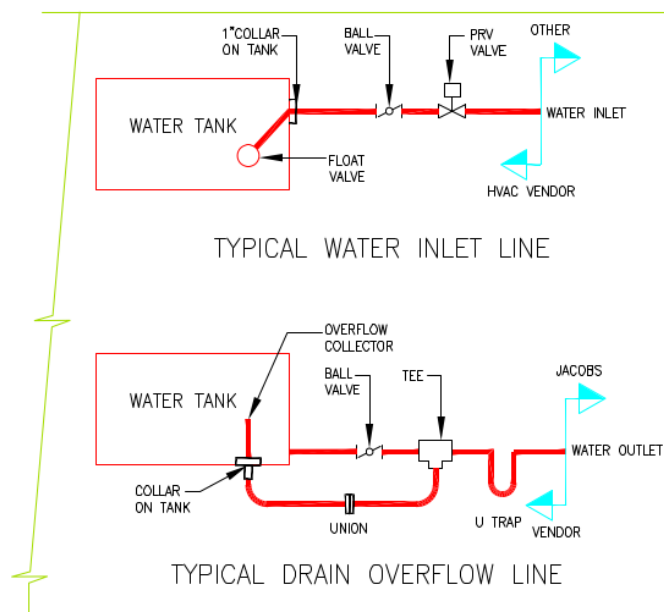
- 2-Way Pressure independent balancing cum Control Valve
- 5 Isolation Valve
- Y-Strainer
- 2 Pressure Gauge at Inlet & Outlet.
- 2 Temperature Gauge at Inlet & Outlet.
- Air Vent
- Drain Valve
- Test Point

Typical Manifold for AHU



TYPICAL MANIFOLD DETAILS FOR CHILLED WATER COIL

Typical Manifold for Direct Evaporative Cooling Unit



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8.5 Valves

HVAC chilled manifold will consist of the following valves.

- Ball valves will be flange end connections and required flanges.
- Butterfly valves will be with flange end connection and required flanges.
- 2-way Pressure independent balancing cum control valve.
- Strainer at inlet of Coil.

Ball Valve and Butterfly valve shall be as per Piping Specification RPMS Document No. J22N03-SPC-PA-820-001 _Rev 0A And RPMS-ENGG-SPC-PP-038-PART-I

8.6 Insulation For Chilled Water Pipe and Ducting

The Insulation for Chilled Water pipe and Duct insulation shall be as per Insulation specifications.

Reference Document No.- RPMS-ENGG-SPC-HV-019

8.7 Electrical Cables

All power cables shall be selected as per standard Electrical specifications for Cables.

Electrical Cables specifications & data sheet Shall be as per Electrical document no.- RPMS-ENGG-SPC-EL-022_r0_v1.0

8.8 Electrical Panel

The AHU Electrical Starter panel shall be in scope of HVAC vendor.

Electrical Panel specifications & data sheet Shall be as per Electrical document no.- RPMS-ENGG-SPC-EL-002_r0_v1.0

Standard Specification for LV Motor shall be as per RPMS-ENGG-SPC-EL-004.

Specification for Electrical requirements for Packaged Mechanical Equipment shall be as per RPMS-ENGG-SPC-EL-035.

8.9 VFD

The VFD shall be suitable for variation in speed of the motor for given application. The range of speed variation shall be from 10 % to 100 % of the rated synchronous speed.

It shall be possible to continuously vary the motor speed with the help of DPT. DPT will give signal to DDC controller & from Controller to VFD. DPT shall be installed across AHU Blower section. Or VFD will operate based on the return Air temperature/Room Temperature and PIBCV will operate based on AHU supply temperature. VFD will modulate based on input received from Field devices 0 – 10 V or 5 – 20 mA.

The VFD shall be in Starter control panel located near AHUs.

VFD compartment shall be provided with cooling fan to keep the temperature of the VFD within normal range.

VFD specifications & data sheet Shall be as per Electrical document no.- RPMS-ENGG-SPC-EL-011_r0_v1.0

8.10 Filters

Following filters with efficiency levels shall be considered.

		Filter Grade			Air Filter Selection
Primary Filtration	Medium Efficiency Primary	US	EU	GLOBAL	
		MERV 2 - 4 MERV 5 - 6 MERV 7 - 9	G2 ≥ 65% G3 ≥ 80% G4 ≥ 90% Average Arrestance	ePM Coarse Arrestance	
Filtration for Air Conditioning Systems	Medium	ASHRAE 52.2 - 2017	EN 779:2012	ISO 16890 2016	
		MERV 10 MERV 11 - 12	M5 ≥ 40% M6 ≥ 60%	ISO ePM ₁₀ >50% ISO ePM _{2.5} 50 - 60%	
		MERV 13	F7 ≥ 80%	ISO ePM ₁ 50 - 70%	

8.11 Interlocking

- Chilled water valves will be closed automatically during the AHU OFF condition.
- Exhaust fans linked with dedicated AHU's shall be interlocked with AHU.
- Limit switch of AHU/Blower access door shall be hard wired with electrical panel (AHU should trip when access door is open).
- Central Fire control system and Fire/Smoke Damper interlocking with AHU/DEC.
- Smoke Exhaust fan shall be interlocked with Fire Signal.
- All motorized dampers of Air handling unit shall be interlocked with AHU/DEC.
- Evaporative Cooling Pump Interlock with Humidistat.

8.12 Axial Fans

Fan shall be licensed to bear AMCA seal.

Fan Casing should be provided with Special Designed Integral Straightening Vanes to reduced turbulence provide high performance & low noise level.

Fan motor base support shall be properly secured (locked and sealed) to the fan housing and be of adjustable type to have precise control of motor shaft central position as well as running clearance between blade tips and casing. Motor (KW/HP) shall be able to be changed or upgraded at site without changing fan housing or ducting construction.

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Fans supplied shall be complete with factory fabricated mounting bracket (ceiling or foot mounted) and suction/discharge matching flanges as accessories.

Fan casing Shall be Epoxy coated mild steel/Galvanized Steel. Aero foil blades are made of Aluminium Alloy.

All hubs shall be Aluminum alloy and provide self-balancing, anti-static, anti-sparking characteristic is preferable.

Impellers shall be secured to the drive shaft by a key and keyway.

Fans shall not exceed 1500 RPM.

All fans after assembly shall be dynamically trim-balanced to ISO1940 and AMCA 204/3 - G2.5 quality grade. A computer printout with vibration spectrum analysis shall be attached to the fans.

The Fan should be AMCA Certified for Air & Sound Performance.

Electric motors shall comply be IE-3 type and comply Standard Specification for LV Motor shall be as per **RPMS-ENGG-SPC-EL-004**

Floor mounted / ceiling suspended supporting arrangement with complete structure (to be approved by consultant during execution) to be included in offer. Structural design shall be approved by consultant.

All fans shall be clearly labelled before dispatch, indicating unit reference and duty.

The Contractor shall advise his standard painting procedure for prior approval.

Anti-vibration Floor mount spring isolator and ceiling hung spring isolators shall be provided for each fan. These shall be selected to be suitable for the working weight and speed of the fan.

Guaranteed sound power and pressure levels for fans shall be provided before approval for manufacture can be given.

The external noise (SPL) from a fan shall not exceed 65 dB (A) at 1m free field distance from the equipment.

Flameproof axial fan wherever specified shall be provided with flameproof motor and spark proof blades / body. For smoke exhaust application, axial fan shall be capable of operating at 250° for 2 Hrs and Smoke Fan motor shall be with H Class Insulation.

8.13 Motor Name Plate Requirement

Labels shall be of SS and shall be securely fixed to the external casing of the unit. Nameplates shall show as a minimum:

- Manufacturers Name and Serial Number
- Type and Duty
- Rated kW
- Motor Frame Size

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- RPM
- Voltage Rating
- Current Rating
- Insulation Class
- Frequency
- IP Rating
- Equipment Tag Number
- Weight
- Air Flow Direction
- Air Flow Capacity
- Static Pressure

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9. System Design Parameters

Design Parameters	
Chilled water entering temperature for Cooling coil of Comfort area AHU	7 Deg C
Chilled water Return temperature for Cooling coil of Comfort area AHU	12 Deg C
Face velocity across filter & coil	NMT 500 FPM
Type of Motor and Fan for - Comfort area AHU - DEC	- Plug type fan with IE3 motor compatible with VFD. - DIDW Centrifugal fan with VFD.
Maximum Motor speed	2880 RPM (for 2 Pole) 1440 RPM (for 4 Pole) 970 RPM (for 6 Pole)
Motor Frequency	50 Hz
Maximum Fan frequency (during dirty condition)	49 HZ
Face velocity across return air grill	NMT 500 FPM
Maximum Duct Velocity of Main Duct for Comfort area	0.06 to 0.08 inch/100ft or 1500 FPM to 2000 FPM
Maximum water velocity in Chilled Water manifold piping	1.5 to 2 m/s
Maximum water pressure drops in Chilled Water manifold piping	4 ft / 100 r. ft of water column
Pressure drops across control valve	0.5 Kg/cm ²
Pressure drops across 7-12 Deg.C Cooling coil	Max 5 meter of H2O

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10. Smoke Extraction System

The Smoke extraction system unit will be provided for Ingot process as the factory comes under medium hazard building as it is exceeding 750 sq. M as per NBC 2016 Part-4 ANNEX M (Clause 6.7.3) FIRE PROTECTION CONSIDERATIONS FOR VENTING IN INDUSTRIAL BUILDINGS.

The extraction of the smoke in case of fire will be done through axial fan installed at corridor both side (east side & west side). The capacity of Smoke exhaust system will be designed at 4 Air changes of the total effective volume of Process area.

Smoke exhaust fans will be common for multiple areas and will have ducting if required and CSFD for separating the area. Fan selection shall be max. air qty of a single area.

All smoke exhaust ceiling fans-local on/off switch to be provided for maintenance purpose.

For life safety equipment power supply design (Manual override electrical provision should be provided at the fire tender entrance lobby/fire control room of the building.

The effective volume of Process area = Total Volume of Process area – Machine Volume of Process area.

Operation of Smoke Extraction Tube Axial Fans

Smoke Extraction Fans shall be supplied with Motorized Dampers/Gravity Dampers as applicable. During Normal operation Smoke Extraction Fans shall be “OFF” and Motorized Dampers/Gravity Damper shall be closed.

In case of fire, based on signal from Fire Alarm System, Smoke Extraction Fans shall be “ON”, and Motorized fire Dampers shall be open.

Smoke Extraction Fans and associated Motorized Fire Dampers shall be field by fire.

The Tube axial fan shall operate on 250 Deg.C, Class H & 2-hour fire rating.

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11. Pressurization System for Fire Escape Corridor

Pressurization Fan shall be provided for Fire escape corridor. As per NBC, these Fire Escape Corridor shall be pressurized with 25 Pa & 30 Pa & air volume shall be calculated on based on this pressure requirement.

Operation of Pressurization Fans

Pressurization Fan shall be provided for Fire escape corridor. In normal case 50% capacity shall be working. In case of fire signal 100% capacity shall run once the signal received from F & G panel

Pressurization of Staircases and Lift Lobbies (If applicable as NBC2016)

Staircase and lift lobby pressurization is a method adopted for protecting the exits from ingress of smoke. Air is injected to raise the pressure slightly above the pressure in adjacent parts of the building preventing ingress of smoke during Fire scenarios. Pressurization is achieved by providing dedicated pressurization fan. Quantity of outside air intake is calculated based on the pressure parameters mentioned below, as per NBC 2016.

Pressure difference for staircases: 50 Pa

Pressure differences for Lobby: 25-30 Pa

The operation of the stairway pressurization system needs to be interlocked with that of the emergency management system serving this building.

The actual capacity of the stairway pressurization system to be determined during the detail design.

12. Integration Between HVAC and Building Automation System (BAS)

DDC controller for individual system for HVAC Control shall be provided. Provision of Hook-up all DDC controllers will be with centralized BMS system (building wise) further interface of BAS to clients, SCADA system shall be client's scope.

12.1 AHU

The following Parameters will be taken into considerations from individual Systems for BAS Connectivity:

- AHU ON/OFF Command
- AHU Auto/manual status
- AHU Run Status
- AHU Trip Status
- Push buttons start / stop status.
- Supply & Return Fire Damper Open/close status
- Motorized damper on / off Command.
- CHW Control Valve Command from AHU Supply Air temperature
- Supply Air Temp
- Return Air Temp
- Differential Pressure Transmitter Signal for Filter
- VFD Command and Mapping/integration to BMS/DDC
- Duct Mounted T and RH Sensor
- Fire Signal from Main central Fire Control Panel

12.2 DEC

The following Parameters will be taken into considerations from individual Systems for BAS Connectivity:

- DEC ON/OFF Command
- DEC Auto/manual status
- DEC Run Status
- DEC Trip Status
- DPT Signal for Filter
- Push buttons start / stop status.

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- Motorized Fire damper on / off Command.
- Pump ON/OFF Status with humidistat placed inside serving Area.
- Supply & Return Fire Damper Open/close status
- VFD Command with mapping/integration to BMS/DDC.
- Fire Signal from Main central Fire Control Panel
- Water level Sensor feedback.

12.3 VRF

- VRF ON/OFF Command
- VRF Run Status
- VRF Trip Status
- T and RH Sensor
- Fire Signal from Main central Fire Control Panel

12.4 General Points

- All the selection of field devices such as Temperature Sensors, Relative Humidity sensors, DPT etc shall be as per BMS specifications.
- Individual/Combined AHU/DEC will have its dedicated DDC Controller for controlling of field devices such as DPT, Temperature-RH sensors, Chilled water control valve actuator, Motorized Damper Actuator, VFD, Electrical Panel.
- Field devices such as Temperature-RH sensor will be used to modulate chilled water.
- For Instrumentation and Control kindly refer attached RPMS MR documents as Applicable.

Control & Instrumentation requirements of BMS for HVAC system shall be as per Doc No. J22N03-JDB-EWE305-820-001 (Standard Specification for Programmable Logic for Controller), Doc No. J22N03-SPC-JL0001-820-001 (Design Basis for Control and Instrumentation) & Doc No. RPMS-ENGG-SPC-CI-021 (Standard Specification for BMS)

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13. Vendor Document Requirement

Vendor shall submit Design / Functional Specifications, Piping, and Instrumentation Diagram (P&ID), General Arrangement drawings, HVAC Shop Drawings, Component List, Instrument List, Control and Electrical Drawings, Equipment datasheets and performance curves, Calibration and test certificates, Loading Data, or any other drawing /document required for installation, commissioning, operation, and qualification of system. The drawings / documents shall be provided as hard copies and wherever possible as electronic version. The supplier shall provide documentation reflecting “as built” condition with final delivery. The supplier shall specifically provide following documents & not limited to

- HVAC Equipment and Accessories Datasheets
- Test Certificates of all HVAC Equipment and accessories
- Datasheet of all HVAC Equipment and accessories
- Fan Curve
- Performance curve of Equipment as Applicable
- Equipment software selections
- QAP & ITP of all HVAC Equipment.
- GA of all HVAC Equipment and accessories
- GA drawing / wiring drawing/ single line diagram of all electrical panels
- Datasheets and GA drawings of Control system components
- Datasheets and GA drawings of all field instruments
- Detailed I/O summary and system architecture drawings, Control logic write up.
- Motor Data sheets & Curves.
- Coil Selection
- Catalogue of all bought out items.
- Calibration Certificate
- Design Qualification, Installation Qualification, Operational Qualification, Performance Qualification.
- Inspection Test Plan.
- Room Air balancing Report.
- Pressure differential across filters test record.
- Room Relative humidity Test Record.
- Room Temperature Test Record.

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14. Acceptance Test

The Individual Equipment's performance will be performed at Vendor's factory (FAT) & Client's site (SAT) as applicable.

FAT

The Contractor shall perform all inspection and tests of the system as a whole and of components individually as required, under the supervision of the Client / Jacobs, in accordance with the provisions of the applicable ASHARE or approved equal, guideline requirements, applicable standards, as instructed by Jacobs / client and as per load requirements. Vendor to Submit ITP based on categorization plan provided in MR and get it Approved by Jacobs/RIL. Vendor to demonstrate equipment testing as per Approved ITP in presence of TPI/Jacobs/RIL.

SAT

Contractor shall perform Site acceptance test based on installed system as a whole as per design requirement.

- Vendor to Submit ITP as per the categorization plan attached in the MR.
- Vendor to provide SAT as per design requirement.

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15. Assumptions/ other Considerations/ Assumptions/ Calibration Required

- Operation of the HVAC system shall be considered as 24 hours per day and 7 days a week.
- All Air-conditioned areas of plant shall run simultaneously.
- Fire Damper will be installed in supply, return air ducts & crossing fire rated walls as per applicable code-NBC, NFPA etc.
- All motors shall be considered with efficiency class of IE3 as applicable.
- AHU Drain piping up to nearest drain trap shall be in HVAC vendor scope.
- For AHU / Ventilation unit motor kW Estimation shall be as per Static Pressure calculation in dirty / clogged condition.
- R&D /Lab Area have considered as Unclassified.
- Lifting lugs shall be supplied for bulky and heavy items.
- Roof and Wall Load for Attic area is not considered for Puller Area.
- Envelop load for Process areas DEC calculations considered up to 3-meter height for occupied/operational zone as the hot air is exhausted from room upper zone to atmosphere.
- HVAC equipment drainpipe from equipment to nearest drain trap shall be in HVAC Contractors scope. U-trap will be of SS-304 material that will be provided by HVAC equipment vendor.
- VFD will have in-built controller for controlling of HVAC field devices like DPT, T-RH sensors, Chilled water actuator and Electrical Heater. Input output summary of individual AHUs will be provided during detail engineering to select appropriate controller.
- For AHU / Ventilation unit motor kW Estimation shall be as per Static Pressure calculation in dirty / clogged condition.
- The equipment shall be completely shop assembled to check fit prior to shipment. All critical dimensions shall be verified, and all motors checked and tested in the shop.
- Heat Load Estimation will be done based on room wise parameters or Room book.

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16. Energy

The following measures have been considered for energy conservation.

- To highly provide energy friendly HVAC system for all the AHUs, AHU's shall be designed with recirculation mode (once thru mode wherever applicable in emergency for LEL sensor activation) complete with mitigation of risks (recirculation mode shall be selected wherever permissible based on process requirements).
- DPT across all filters to monitor choked conditions.
- Modulating type control valve shall be considered as per load requirement.
- Actual equipment heat dissipation for optimized & precise design of HVAC system.
- Use of energy efficient lights to reduce cooling capacity requirements.
- Controlling the conditioned air leakages helps in optimized use of HVAC system & saving the power consumed.
- Optimized designing of ducts & pipes routes reduces the transmission losses.
- back-up supply from DG will be provided for critical areas. critical areas to be decided with client. Adequate insulation thickness to avoid increase temperature rise and condensation control.

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17. Safety

- Fire dampers shall be installed at AHU supply & return side.
- Electric panels shall be properly grounded with no unsecured joints.
- HVAC system shall be provided with suitable arrangement to acknowledge alarms.
- Electrical installation shall be following local code of practice.
- AHU to trip in case access door is open. Door limit switches to be considered.
- Mechanical guard for all rotating parts.
- Emergency push button to be provided in the electrical panel.
- Fire dampers shall be provided for ducting crossing one fire compartment to other fire compartment to avoid fire spread from one area to other area.
- Smoke sensor shall be installed in return air duct to stop AHU in case of fire. Use of Fire-Retardant Class 'O' rated insulation both for ducts & pipes
- remote start-stop push-button/Isolator is required for all HVAC equipment placed away from its electrical panel.

18. Abbreviations:

- ACPH : Air Change Per Hour.
- AHUs : Air Handling Units
- DEC : Direct Evaporative Cooling Unit
- ASHRAE : American Society of Heating, Refrigerating & air conditioning Engineers.
- BMS : Building Management System
- cGMP : Current Good Manufacturing Practice
- DBT : Dry bulb Temperature
- DDC : Direct Digital Controller
- DQ : Design Qualification
- EU : Exhaust Unit
- FF : First Floor
- FFU : Fan Filter Unit
- GMP : Good Manufacturing Practice
- HEPA : High Efficiency Particulate Air Filter
- HVAC : Heating, Ventilation & Air conditioning
- ISHRAE : Indian Society of Heating, Refrigerating & air conditioning Engineers.
- MCC : Motor Control Centre
- NC : Not controlled
- NMT : Not More Than
- RH : Relative Humidity
- SMACNA : Sheet Metal and Air-conditioning Contractors National Associates
- SOP : Standard Operating Procedure
- TR : Ton of Refrigeration
- VCD : Volume Control Damper
- VFD : Variable Frequency Drive
- VU : Ventilation Unit
- WBT : Wet Bulb Temperature